

CHAPTER 11

INTRODUCTION TO DAMPING TREATMENTS

Dynamic responses of machines and structures are primarily governed by three properties - mass, stiffness and damping. The first two of these are most often readily understood by the vibration analyst. However, the third property, damping, is probably the most elusive and seemingly complex property of these three and is definitely the most difficult to predict. Unfortunately, the word "damping" has very loosely been used for years to apply to any number of vibration and noise reduction problems. That is, this term (damping) is frequently used with almost self-contradictory phrases such as "damp out the sound in a room", or "we use damping mounts under our machines". This indicates a state of popular confusion between the mechanisms of vibration damping, vibration isolation and sound absorption. Therefore, the first item we must do is to clearly define what is meant by "damping":

- **Damping** - Damping is the dissipation of vibratory energy with motion or with time, converting this energy from mechanical motions to other types of energy, chiefly heat. Therefore, the measurement of damping of a structure or material reveals its ability to rid itself or disperse vibration over time and distance. The lower the damping, the longer it will require a component to come back to rest once it has been impacted by an external force.

Figure 1 helps illustrate the most important function of damping. Figure 1 shows three distinctly different regions (A, B and C) which shows the major limiting factor to vibration in each of these three regions. Region A shows a rotor operating below resonance where vibration due to unbalance will increase with the square of speed. Note that the dominant resistance to vibration in Region A will be the stiffness of the support system where stiffness, K, is expressed in units of pounds per inch (lb/in). Therefore, the amount of damping within the rotor has only negligible influence on resultant vibration when the rotor is operating within Region A.

However, when the rotor accelerates and begins to approach resonance in Region B, the major limiting factor to vibration is the amount of damping within the system (where the "system" can include the rotor, supporting bearings and even the support structure). When the rotor goes into resonance, the only limiting factor to infinite vibration amplitude is the amount of damping within the system. In fact, lightly damped systems can easily experience from 20 to 100X the amount of vibration when at resonance, again limited only by the amount of damping. These reduced vibrations due to greater damping in turn will result in lowering of stress, fatigue and sound radiation.

Figure 1 next shows that if the rotor is accelerated beyond the resonant amplification Region B, it will pass into Region C where both system damping and stiffness begin to have less and less influence on amplitude response. When fully into Region C, the rotor will respond almost as a pure mass, with almost no resistance supplied from spring stiffness or from damping.

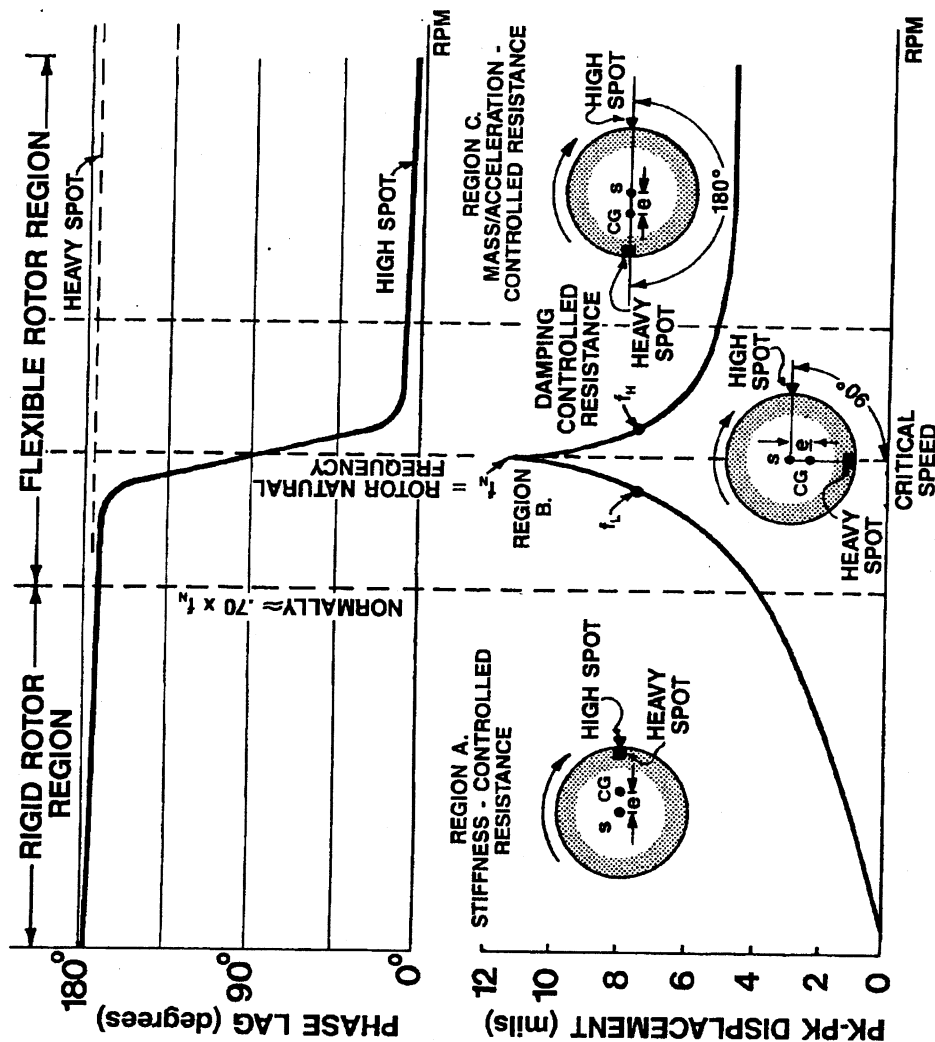


FIGURE 1
CHANGE OF VIBRATION DISPLACEMENT AND PHASE LAG
WITH RPM ABOVE, BELOW, AND AT ROTOR RESONANCE (Ref. 1)

$$\frac{X_0}{X_{ST}} = \frac{1}{\sqrt{\left[1 - \left(\frac{f_i}{f_n}\right)^2\right]^2 + \left[2d\frac{f_i}{f_n}\right]^2}}$$

When Forcing Freq. = Natl. Freq. ($f_i = f_n$):

$$\frac{X_0}{X_{ST}} = \frac{1}{2d} = Q = \frac{f_n}{(f_n - f_i)}$$

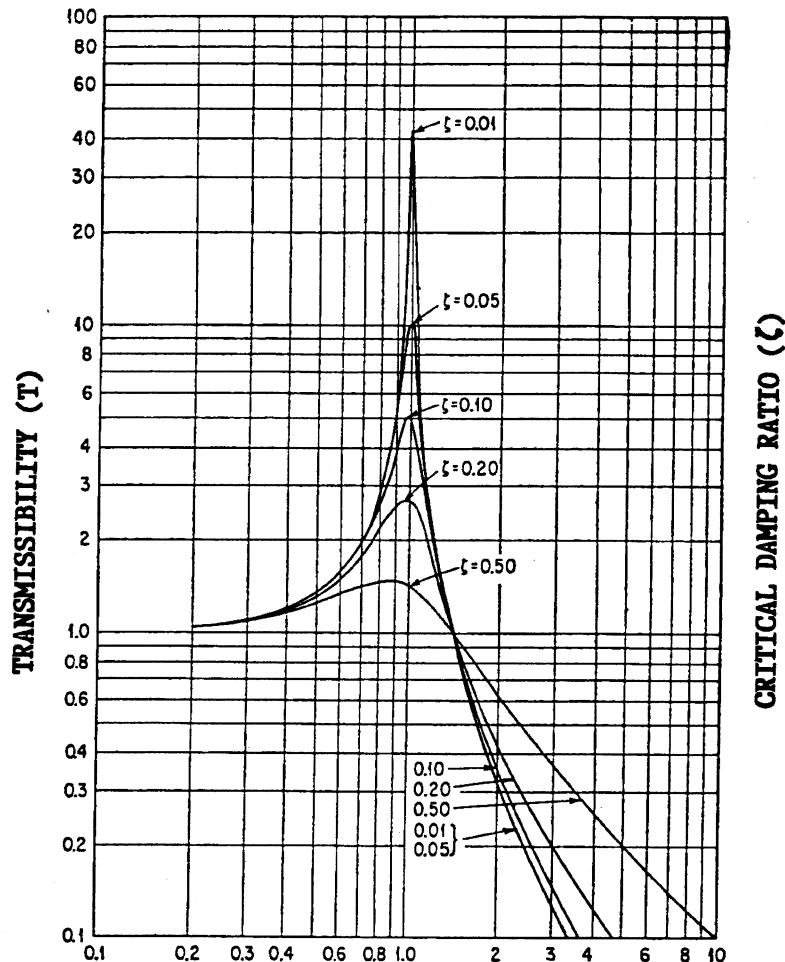
WHERE:

X_0 = Dynamic Deflection at Frequency f_i (mils)
 X_{ST} = Static Deflection (mils)
 f_i = Forcing Frequency (CPM)
 f_n = Natural Frequency (CPM)
 d = Damping Factor (lb-sec/in)
 Q = Resonant Amplification Factor
 f_h = .707 A_n = Upper Half-Power Freq. (CPM)
 f_l = .707 A_n = Lower Half-Power Freq. (CPM)
 A_n = Amplitude of Natural Frequency (mils)

LEGEND

CG = Rotor Center of Gravity
 S = Shaft Geometric Centerline
 e = Eccentricity (in.)
 Heavy Spot = Actual Unbalance Location
 High Spot = Point of Maximum Rotor Deflection

FIGURE 2
THEORETICAL TRANSMISSIBILITY CURVE FOR A RANGE OF DAMPING
FACTORS (ζ) IN A SINGLE DEGREE-OF-FREEDOM SYSTEM (Ref. 2)



$$\frac{f_d}{f_n} = \frac{\text{Driving Frequency}}{\text{Natural Frequency}}$$

$$\text{TRANSMISSIBILITY} = T = \frac{1 + \left[(2\zeta) \left(\frac{f_d}{f_n} \right) \right]^2}{\sqrt{\left[1 - \left(\frac{f_d}{f_n} \right)^2 \right]^2 + \left[(2\zeta) \left(\frac{f_d}{f_n} \right) \right]^2}}$$

Where:

T = Transmissibility

f_d = Driving or Forcing Frequency (Hz or CPM)

f_n = Natural Frequency (Hz or CPM)

$\zeta = c/c_c$ = Critical Damping Ratio = $\frac{1}{2Q}$

Q = Resonant Amplification Factor = $\frac{1}{2\zeta}$

Figure 2 shows how a machine or structure is effected by the amount of damping when operating at resonance. The vertical scale labeled "Transmissibility" (T) describes how much vibration is amplified when operating below, at resonance and when passing through resonance. The horizontal scale is the ratio of the driving frequency (f_d) divided by the natural frequency (f_n). Therefore, when the driving frequency equals the natural frequency, this is the condition of resonance. The equation given on Figure 2 for transmissibility (T) includes the damping term known as the "critical damping ratio (ζ)" which is also sometimes referred to as the "damping factor". Note on Figure 2 that the resonant amplification factor (Q) is also shown and equals 1 divided by 2X the damping factor (see Equation 1). This is proven by looking back at the transmissibility equation and setting the driving frequency (f_d) equal to the natural frequency (f_n). Therefore, when at resonance, the driving frequency coincides with the natural frequency. This driving frequency might be the operating speed, but can be any frequency which coincides with any natural frequency. That is, it might be the blade pass frequency (#blades X RPM) of a pump which lines up with a natural frequency; or a gear mesh frequency (#teeth X RPM), or simply 2X or 3X RPM. If any of these frequencies happen to line up with a system natural frequency, resonance will occur and vibration will be amplified. Typically, there is less and less energy with higher and higher multiples of operating speed so that the resonant vibration response is lower and lower as the higher harmonics of running speed pass through resonance. The important fact here is that numerous forcing frequencies can coincide with a natural frequency to cause resonance (not just operating speed frequency).

Equation 1 helps demonstrate the various forms used to express the amount of damping and shows how it can be calculated. Equation 1 also shows how damping relates to the resonant amplification factor (Q):

$$Q = \frac{1}{2\zeta} = \frac{1}{\eta} = \frac{\pi}{\delta} = \frac{f_n}{f_H - f_L} \quad \text{EQUATION 1}$$

where:

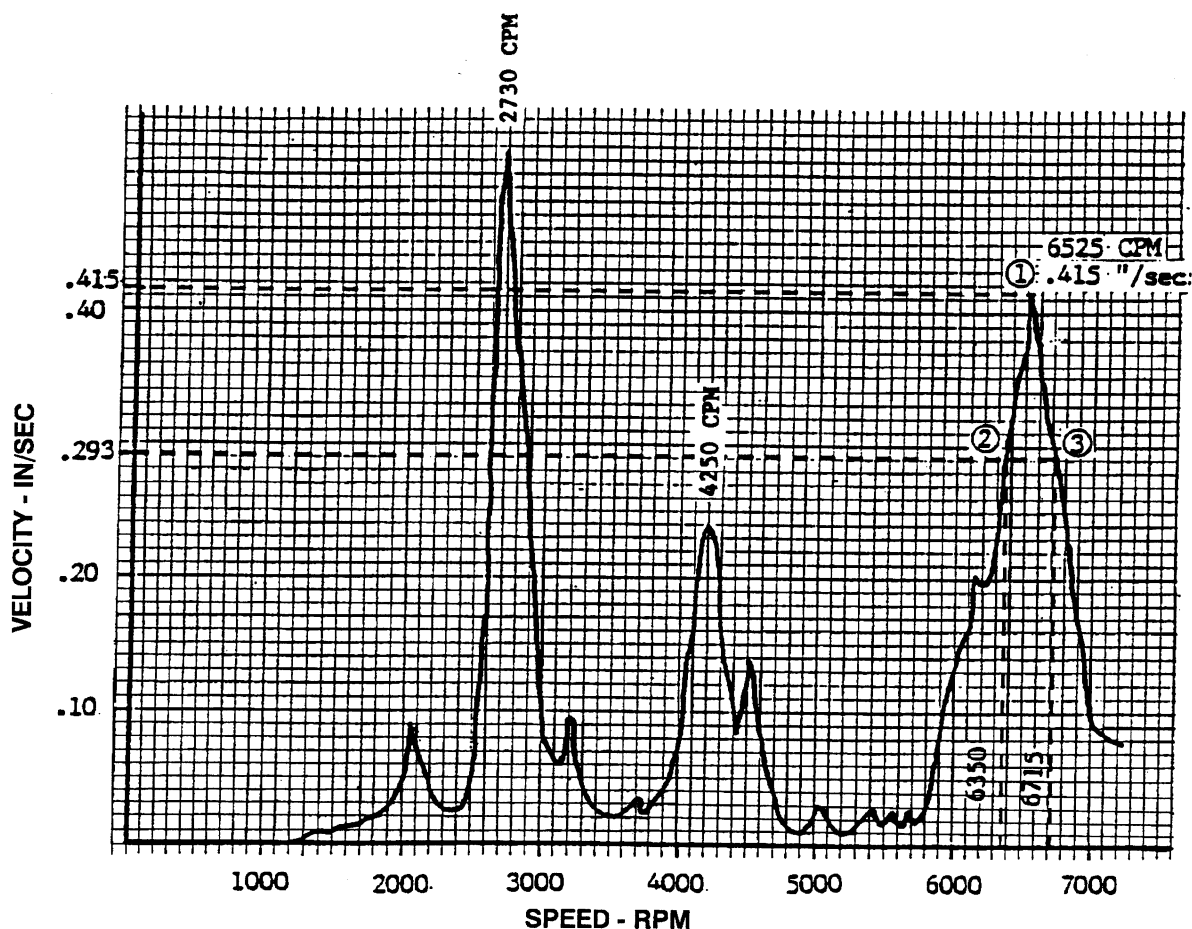
- Q = Resonant Amplification Factor
- ζ = Critical Damping Ratio = c/c_c where c is the viscous damping coefficient (lb-sec/in) and c_c is the critical damping coefficient, or the smallest amount of damping required to return the system to its rest position without vacillation (lb-sec/in).
- η = 2ζ = Loss Factor for Damping Material which is the ratio of energy dissipated per radian to the maximum potential energy. It is a measure of the capacity of a vibration damping material to dissipate vibration.
- δ = Log Decrement Damping Factor which is a measure of the amount of damping based on the rate of decay of free oscillatory motion.
- f_n = Natural Frequency (Hz or CPM)
- f_H = Higher Half-Power Frequency just above f_n with an amplitude equal to $.707X f_n$ amplitude (Hz or CPM)
- f_L = Lower Half-Power Frequency just below f_n with an amplitude equal to $.707X f_n$ amplitude (Hz or CPM)

The importance of the resonant amplification factor (Q) is that Q shows how much higher the vibration response is since the machine is at resonance (correspondingly, it also shows how much lower the vibration could be if the machine were not operating at resonance at this particular frequency). To clarify this, Figure 3 is an actual spectrum captured when attempting to bring up a variable-speed machine to full operating speed (the machine was "supposed" to operate successfully up to 8000 RPM, but did not at that time). Looking at Figure 3, note there

were several regions of higher vibration response including those at 2730 CPM (45.4 Hz), 4250 CPM (70.8 Hz) and 6525 CPM (108.75 Hz). What Figure 3 showed was that operation of the machine either close to 2730 or 6525 RPM would result in high vibration response. Suppose it was desired to operate the machine at 6525 RPM. Figure 3 shows that it would be subjected to a velocity of .415 in/sec which was clearly excessive for this particular machine. Therefore, a calculation of the resonant amplification factor (Q) as well as the damping ratio (ζ) was made on Figure 3. It used the items in Equation (1) to calculate Q where the natural frequency (f_n) was 6525 CPM; the lower half-power frequency (f_l) was 6350 CPM where the amplitude was 70.7% of that at f_n which is the definition of a half-power point; and the higher half-power frequency (f_h) was at 6725 CPM. Then, Figure 3 shows that the amplification factor (Q) was found to be 17.9. This means that resonant amplification was 17.9X higher than it would be if the machine were not resonant at this frequency (in other words, if this natural frequency could be changed by altering mass or stiffness moving it away from 6525 CPM, the resulting amplitude could be almost 18X lower, or .023 in/sec). Figure 3 then shows the calculation of the damping ratio (ζ) which was found to be .056.

Figure 3 helps make two other important points - (1) machines and structures commonly have several natural frequencies (in fact, they most always have a different set of natural frequencies in the horizontal, vertical and axial directions; as well as in each of three "rocking" directions); and (2) each natural frequency has its own specific damping factor (ζ). Looking back at Figure 3, note how much higher the response was at 2730 CPM than it was at 4250 (and likewise higher than at 6525 CPM). Also, note how much wider the resonant response curve was at 6525 CPM than it was at 2730 CPM. This means that the 2730 CPM resonance had a lower damping factor than that at 6525 CPM; and therefore, had higher resonant vibration response.

FIGURE 3
DAMPING RATIO FROM COASTDOWN VIBRATION



- 1 - At peak amplitude of .415 "/sec the frequency is 6525 CPM
- 2 - The half power points are .415 x .707 = .293 "/sec.
- 3 - The frequencies at the half power points are 6350 CPM and 6715 CPM
- 4 - The amplification

$$Q = \frac{6525}{6715 - 6350} = 17.9$$

- 5 - The damping ratio:

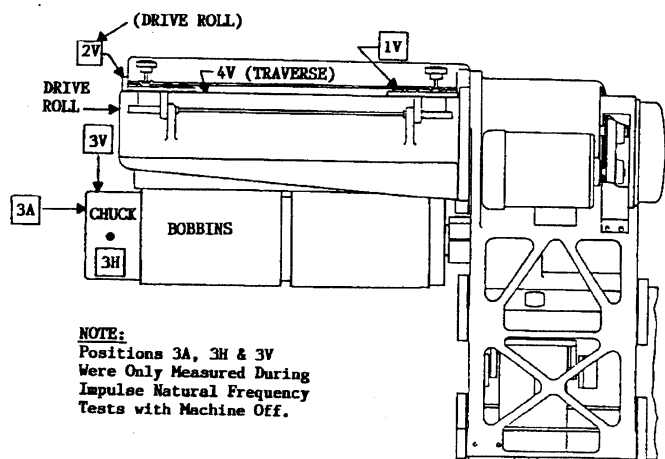
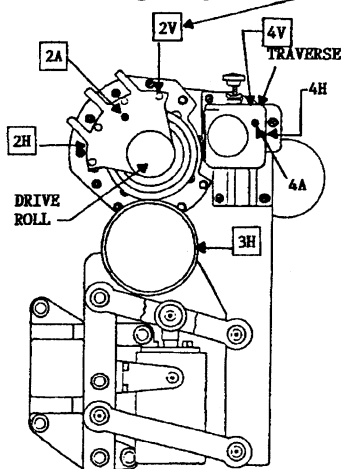
$$Q = \frac{1}{\sqrt{[1 - (\frac{\omega}{\omega_n})^2]^2 + (\frac{\zeta}{C_c} \frac{\omega}{\omega_n})^2}} \quad \text{at resonance} \quad Q = \frac{1}{2 \frac{\zeta}{C_c} \frac{\omega}{\omega_n}}$$

$$\zeta = \frac{C}{C_c} = \frac{1}{2Q} = \frac{1}{2(17.9)} = .056$$

Of course, resonance is something which should be avoided by all machines or structures. However, some machines cannot avoid operation at resonance. For example, there are many machines such as rolling mills or winders which commonly are forced to pass through several resonances during operation. For example, note the synthetic fibers winder shown in Figure 4 which must maintain a constant surface speed (ft/min or m/min) throughout the winding process. Note that the winder is made up of a drive roll which comes down and contacts the chuck spindle which normally will carry anywhere from 2 to 4 bobbins on which synthetic yarn filament is wound. While the drive roll maintains a constant rotating speed throughout the winding operation, the chuck spindle carrying the bobbins constantly decreases in speed as more and more yarn is wound up on the bobbins. Therefore, most all of these types of machines are forced to run through one or more natural frequencies as not only the chuck spindle's speed (RPM) decreases, but also the spindle mass increases up to a full package which commonly might range from 20 to 30 pounds per bobbin. In these cases, the chuck spindle may initially pass through a vertical natural frequency when it is roughly half full; then it might pass through a horizontal natural frequency when it slows down when approaching full package size. Unfortunately, with increasing package size, not only might the chuck spindle be forced into resonance, but it very likely will have higher and higher package eccentricity as it winds more and more product (even though intense measures are made to prevent the filament from laying one strand on top of one another in a process called "ribbon breaking" in the synthetic fiber industry). Therefore, these machines will often suffer excessive vibration when they pass through these resonances, particularly if this occurs near full package size.

In the case of machines like that shown in Figure 4 which must pass through one or more resonances often suffering excessive vibration, corrective measures must be taken to help reduce the amount of resonant vibration response. As this section will show, this can be accomplished by increasing the damping of the system by use of a variety of damping treatments.

NOTE: This Position was Chosen for Spectral Measurements During Package Builds.



NOTE:
Positions 3A, 3H & 3V
Were Only Measured During
Impulse Natural Frequency
Tests with Machine Off.

FIGURE 4
MEASUREMENT LOCATIONS FOR SYNTHETIC FIBERS WINDER

- Types of Damping Treatments** - There are two major types of surface damping treatments. These include the free-layer and the constrained-layer type. Figure 5 shows the free-layer treatment which consists of a single layer of viscoelastic material applied to the base structure, and depends on bending vibration of the base structure onto which it is applied to induce deformation of the viscoelastic material. That is, it depends on the metal on which it is placed to extend and to compress it with each vibration cycle. In so doing, energy is dissipated as a result of extension and compression of the material under flexural stress from the base structure. In the case of Figure 5, note that a constant thickness damping material (3/16 in) was applied to each of five different steel plates ranging in thickness from 1/32 inch up to 1/2 inch thick. Figure 5 shows how this effected the system loss factor (η) which is defined by Equation (1). Figure 5 shows that the damping treatment had greater and greater effect with decreasing steel plate thickness. Figure 5 also shows that there was an optimum temperature where the loss factor was highest, and that it changed from one steel plate thickness to another. This is common with damping materials. In fact, when choosing damping treatments, a whole series of treatment types can be used.

Figure 6 shows performance data for a number of damping materials provided by various vendors. Note how they each have their own set of curves graphing the damping loss factor (η) versus frequency. Notice how much their damping capabilities can vary with temperature. For example looking at one of the damping treatments labelled EAR C-2003, this material is nominally effective at the frequency of 400 Hz (24,000 CPM) when the temperature of the surface onto which the material has been applied is either roughly 40°F to 60°F and at 140°F to 160°F (where the loss factor, $\eta = .60$). It would provide maximum damping at temperatures of 75° to 100°F (where the loss factor = 1.0). On the other hand, this material would provide little damping at temperatures either below 25°F or above 160°F when the loss factor fell below .20. For high temperature surfaces of about 250°F at this 400 Hz frequency, Figure 6 shows a better choice would likely be the Soundcoat DYAD 609.

In general, to provide optimum damping, one must first determine the predominant frequencies present. Then, he must determine the appropriate temperatures to which the damping treatments will be subjected. Once he knows this, by using a chart like that shown in Figure 6, he should then select the optimum damping treatment under these operating conditions. Another important point is that when placing the damping treatments on the structure, they should be located where the vibration displacement is highest in order to be most effective.

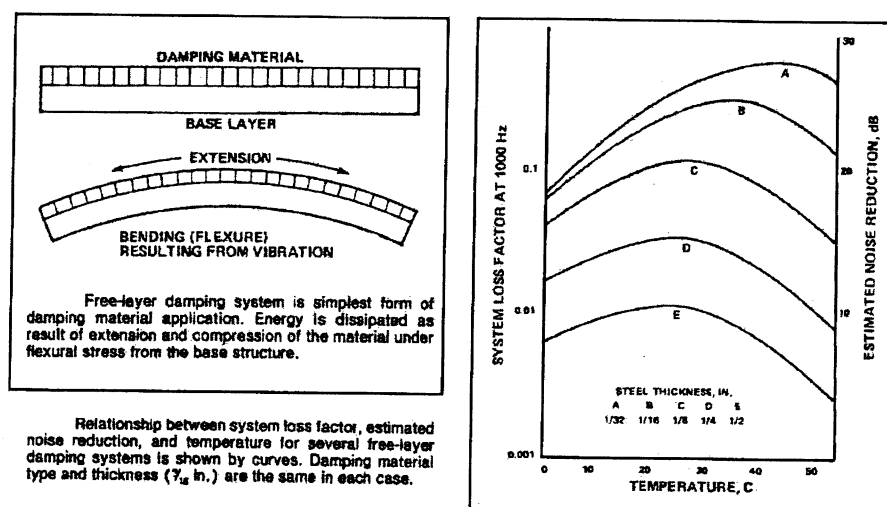


FIGURE 5
FREE LAYER DAMPING (Ref. 3)

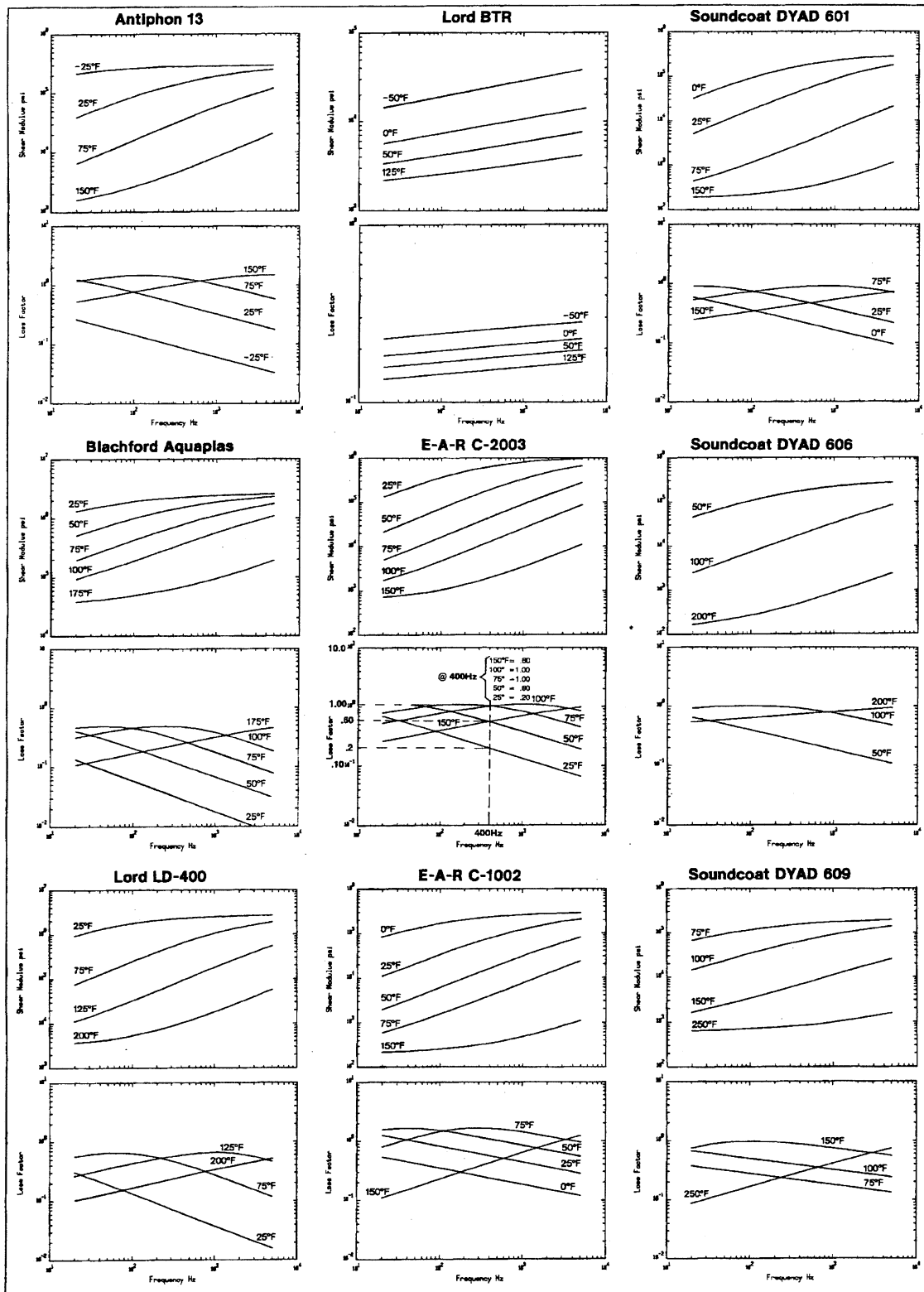
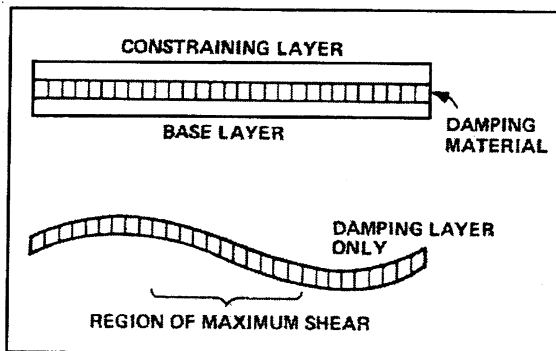


FIGURE 6
PHYSICAL PROPERTIES OF VARIOUS COMMERCIAL DAMPING MATERIALS
 (From Anatrol Corp. materials properties data base.) (Ref. 4)

Figure 7 shows the other major type of damping called "constrained-layer damping". In this case, the constrained-layer treatment consists of two layers - one being a layer of viscoelastic material, along with a layer of relatively stiff material (usually sheet metal). Each of these are bonded to the machine surface. Now, the bending vibration of the base structure induces shear deformation of the viscoelastic material due to the constraint provided on the outer boundary of the viscoelastic layer of the added layer of steel material. Therefore, one of the key parameters in damping efficiency will be determined by the adhesive bonding of the damping material to the base layer and to the constraining layer. This adhesive must have high shear stiffness. The shear strain in the adhesive will reduce the strain in the damping layer which will reduce its overall effectiveness. In general, constrained-layered damping systems will be much more efficient than free-layer treatments since vibration energy will be dissipated not only through deformation of the damping material, but also of the adhesive in constraining layers. In addition, constrained-layer damping often adds strength to the structure which can help lower vibration even in the regions below the natural frequency since this region will predominantly be stiffness controlled. Here again, Figure 7 shows that the constrained-layer treatments will likewise be temperature dependant having their certain optimum temperature and frequency regions in which they will be most effective.



Constrained-layer damping system is usually recommended for stiff, structural applications. Energy is dissipated as a result of shear deformation of the damping layer.

Relationship between system loss factor, estimated noise reduction, and temperature for several constrained-layer damping systems is shown by curves. Damping material type and thickness ($\frac{3}{16}$ in.) are the same in each case.

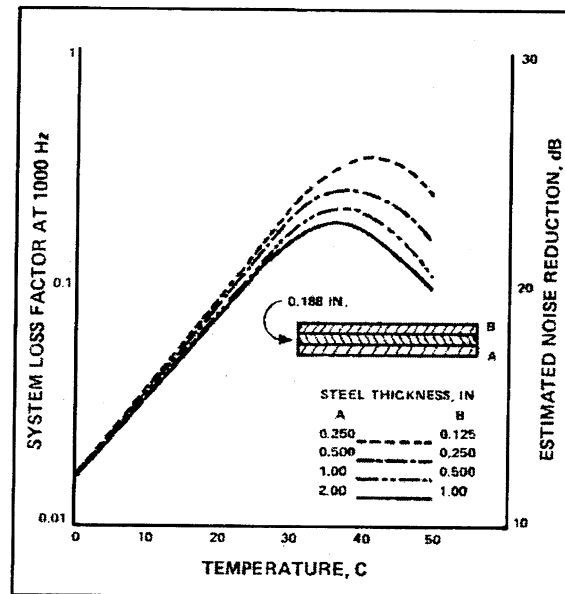


FIGURE 7
CONSTRAINED LAYER DAMPING (Ref. 3) EFFECTS OF TEMPERATURE, FREQUENCY AND THICKNESS ON THE RESULTANT SYSTEM DAMPING

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